Prakul Sunil Hiremath

Applied Science | Machine Learning | LLM Post-Training | Evaluation & Data Quality

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Summary

B.Tech. Computer Science student doing empirical machine learning research on failure modes in LLM post-training and evaluation, with a second focus on data leakage and dataset quality. Two **NeurIPS 2026 accepted posters** (one sole-author). I run instrumented multi-GPU experiments (vLLM, Ray), design the controls and ablations that isolate a cause, audit my own experiments for validity, and release tested open-source Python and C++ research software.

Education

Visvesvaraya Technological University (VTU)

Aug 2023 – Aug 2027 (expected)

B.Tech. Computer Science & Engineering | *CGPA 8.02/10 (six semesters completed)* — Belagavi, India

Status: 7th of 8 semesters (ends Dec 2026); 8th semester remaining.

Coursework: Linear Algebra, Multivariable Calculus, Probability & Statistics, Design & Analysis of Algorithms, Theory of Computation, Artificial Intelligence & Machine Learning, Operating Systems.

Earlier: Pre-University (Science, PCM), Prerna PU College, 91.00% (JEE Main 2023, 84 percentile) | Secondary (CBSE), Billimoria High School, 95.6%.

Publications & Research

Latency-Conditioned Selection Bias in RLHF

NeurIPS 2026, accepted poster (sole author)

Asynchronous RL and PPO training dynamics for reasoning models — Poster Page

- Showed that discarding slow rollouts in asynchronous RLHF acts as an **implicit length filter**, because generation time grows with token count, so long reasoning trajectories are under-sampled. Proved the PPO gradient is attenuated (and can invert in a minimal MDP) when $\mathrm{Cov}(R, \alpha) < 0$; importance-sampling corrections cannot fix this because every retained sample comes from the current policy.
- Instrumented a **vLLM** server (LLaMA-3-8B, 12,000 rollouts): latency explains 91.8% of length variance ($r^2{=}0.918$). In PPO runs (GPT-2, 10 seeds; LLaMA-3-8B on a multi-GPU cluster with Ray, 3 seeds), accuracy fell **68.4%** $\rightarrow$ **44.1%** under truncation. A throughput-matched synchronous baseline (67.8%) and a synthetic latency-injection experiment ruled out batch-size and other confounds.
- Introduced **EGI**, a diagnostic that flagged degradation 600–1,700 steps before reward curves diverged, and **SPW**, an inverse-probability correction. At 8B scale, SPW-trained accuracy reaches **97.8%** of on-policy-trained accuracy (ratio of the two), with $<$0.3% throughput overhead.

When Does Sequential Detection Collapse to a Scalar?

NeurIPS 2026, accepted poster (first author)

Sequential detection under partial observability — Poster Page

- Proved a necessary-and-sufficient condition (Extended Decision Sufficiency, checkable in $O(K^2)$) for when a scalar score is Bayes-optimal in multi-regime detection. When it fails, every scalar detector incurs a positive, computable lead-time loss regardless of architecture or training; also derived a robustness bound of $O(\eta/(1{-}\rho))$.
- Predicted vs. observed lead times agree within 0.4 steps (MAE) on 72 synthetic configurations. On CICIDS2017 the method reaches **14.3 steps** of anticipation vs. $\leq$1.5 for LSTM, TranAD and DeepSAD; a parameter-matched control rules out regularisation as the explanation. Co-authors: P. M. Bagawan, S. Bhekane.

Calibration Drift Under Reasoning

Preprint (with H. R. Hiremath)

LLM reasoning, calibration and adaptive stopping — Paper

- Studied how larger chain-of-thought budgets can make LLMs systematically overconfident. Proposed a Hypothesis Lock-In model and **CABStop**, a calibration-aware rule for when to stop reasoning. Llama-3.1-8B shows non-monotonic calibration; Llama-3.3-70B results are inconclusive and reported as such.

The Hiremath Early Detection (HED) Score: A Measure-Theoretic Evaluation Standard for Temporal Intelligence

Evaluation methodology for time-critical detection — **Preprint** (sole author) | Paper | Code

- Proposed a time-aware detection metric that credits *when* a detector responds, not only whether it is correct (AUROC ignores temporal ordering). Proved three axioms (temporal monotonicity, invariance to pre-attack bias, decomposability into acuity, lead and calibration).
- On NSL-KDD, a switching state-space detector scores 0.0643 vs. 0.0132 for a Random Forest baseline under the metric (block-bootstrap, $p < 0.001$).

Also: *Shared Feature-Based Multitask Model for Faux-Hate Classification in Code-Mixed Text*, ICON 2024 (co-author; Macro-F1 0.7445, 10th in the shared task; ACL Anthology); *Early Detection of Latent Microstructure Regimes in Limit Order Books*, arXiv:2604.20949 (HMM-entropy early-warning detector, $+18.6 \pm 3.2$ steps lead time over 200 simulations).

Under review: *Evaluation Collapse* (ICLR 2027); *Information Is Not Enough* (AISTATS 2027); *Sensitivity–Uncertainty Alignment* (NeurIPS 2026 Evaluations & Datasets Track).

Research Experience

Research Intern, Capability Dynamics

Sep 2026 – Present

ITISMU, Dept. of CSE, VTU (advisor: Dr. M. N. Birje) — Belagavi, India

- Studying whether the initial spectral structure of a one-block Transformer predicts the onset of generalisation (grokking) on modular addition (mod 97). Pilot: 30 seeds $\times$ 20,000 steps in PyTorch.
- Audited my own train/test split and found **commutativity leakage**: at the first checkpoint with $\geq$95% train accuracy, test accuracy was 91.3% on pairs whose reversal was in training and 0.32% on pairs whose reversal was not. Treated the original crossing-time estimates as provisional.
- Reported two further negative results (an early-loss predictor that did not beat the mean baseline; an uninformative permutation control). Designed a corrected pair-orbit split and a pre-specified censored time-to-event analysis; corrected experiments are in progress.

Research Lead, NLP & AI Research Group
Mar 2026 – Aug 2026

Dept. of CSE, VTU (advisors: Dr. Rashmi Rachh, Dr. Sanjana Kavatagi) — Belagavi, India

- Led research on low-resource Indian languages (Kannada, Telugu): hate-speech identification in code-mixed and obfuscated text, data labelling and classification with RoBERTa-based workflows; set up a student research programme in regional-language NLP.

Founder and Research Lead
Feb 2024 – Present

Aliens on Earth (AoE) Research Collective — Belagavi, India

- Lead a student research group on mathematical machine learning, latent dynamical systems and epistemic uncertainty. Group projects have received Rs. 20 Lakhs+ in competitive funding (MSME Idea Hackathon 4.0, Startup Karnataka NAIN 2.0, FKCCI Inno Manthan).
- An AI-assisted algal photobioreactor project was published as a case study in the *Casebook on AI in Energy* at the India AI Impact Summit 2026; uncertainty-aware monitoring work presented at AAC 2025 (Mumbai).

Researcher, AstroSat Data Analysis
Sep 2025 – Feb 2026

Cosmic Calculus & Astronomy Club, VTU (advisor: Shankar Patil, PhD candidate) — Belagavi, India

- Built an unsupervised anomaly-detection pipeline for AstroSat SXT light curves, calibrated with synthetic transient injections into real background data to quantify detection probability and false-trigger rates; presented at the AstroSat conference, U R Rao Satellite Centre, Bengaluru (Jan–Feb 2026).

Open-Source Research Software

Temporal Leaks
May 2026

Look-ahead-bias and leakage detector for time-series ML (Pandas and Polars) — GitHub | PyPI

- Perturbs future data and flags any feature whose past values change. Typed (mypy `-strict`), ruff-clean and CI-tested.
- Case study on the Beijing PM2.5 dataset (42k hourly rows, next-hour forecast): injected 8 look-ahead features (centred windows, global scaling). The audit flagged 8/8 and passed the causal rewrite; an independent truncation-invariance check (recompute features on history cut at T) confirmed 8/8 changed and 0/8 causal features changed. Leaky features with shuffled CV beat a persistence baseline (RMSE 22.5 $\pm$ 1.2 vs. 24.4 $\pm$ 1.7); causal features with walk-forward CV did not (26.0 $\pm$ 4.6 vs. 23.5 $\pm$ 2.6; $\pm$ is std over 5 folds).

Latency Gym
Jun 2026

C++20 / Pybind11 Gymnasium environment for RL latency optimisation — GitHub | Zenodo

- Tail-risk-aware reward (mean + variance + drop penalty); PPO (Stable-Baselines3) trained against a random baseline. Zero-allocation hot loop at $\sim$100 μs/step; 50+ deterministic tests.

hed-score
2026

Reference implementation of the HED metric (Apache-2.0) — GitHub | PyPI | Zenodo

- Axiom-verification tests, NSL-KDD and synthetic-shift experiments, and a reproducible Colab.

Also: ChronosDefrag (pure-PyTorch contrastive regime discovery; 19 deterministic tests).

Technical Skills

Machine Learning & LLMs: PyTorch, Hugging Face Transformers and PEFT, scikit-learn, PPO and policy gradients, RLHF experimentation, SFT, instruction tuning, LoRA, QLoRA, RoBERTa, fastText

Data & Evaluation: NumPy, SciPy, Pandas, Polars, SQL; leakage auditing, calibration and uncertainty, benchmark and metric design, ablations, survival analysis

Systems & Engineering: Python, C++, Bash; vLLM (latency instrumentation), Ray, multi-GPU training, Slurm, Weights & Biases, Docker, Linux, Git, Pytest, GitHub Actions, PyPI packaging

Service, Leadership & Languages

Invited talks: OpenSearchCon Japan 2025 (Tokyo); CommunityOverCode Asia 2025 (Beijing) | **Sponsored delegate:** Linux Foundation Open Source Summit 2025 & 2026

Reviewer: NeurIPS 2026 Workshop NeurReps | **R&D Lead (volunteer):** GDG on Campus Belgaum, Nov 2026 – Present

Mentor (volunteer): Career Buddy, Wadhwani Foundation, Jun 2026 – Present (10+ one-on-one sessions)

Languages: English, Hindi, Marathi, Kannada (fluent); Gujarati, Telugu (understand, limited speaking)